

Response to Reviewers

Applying the Memory Measurement Model (M^3): A Tutorial Using the bmm R Package

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We thank the editor and both reviewers for their careful and constructive feedback. The comments raised substantive and helpful points that we believe have meaningfully improved the tutorial. Below we address each comment in turn. For each point we state our position in the reply, describe the changes made in the manuscript, and — where relevant — note where results from updated analyses inform the revision.

Throughout their examples, the authors demonstrate how to test hypotheses about choice rules and parameter differences between conditions via the Bayes factor. [...] The authors note that "The bmm package sets weakly informative default priors based on known parameter ranges" (p. 14), but neither show or discuss the appropriateness of these priors for their model comparisons. [...] One particular challenge is to specify parameter priors to construct fair model comparisons for non-nested cognitive models. [...] This is relevant to the first example, in which the authors compare two models using different choice rules (simple choice and softmax). These model have the same latent parameters a and c but their mapping to the observed outcomes is different. In both models the same "weakly informative default priors" are used. Without knowing the prior distributions and the implied predictions it is difficult to gauge whether the resulting model comparison is meaningful. [...] Although it was the comparison of decision rules that had me scratching my head, this point is relevant to other model comparisons—especially where priors are not placed directly on the parameters of interest, but on transformations of model parameters. For example, when the authors compare parameters across conditions, they rely on implied priors because they opted for cell means coding rather than placing priors directly on condition differences. It is unclear if the implied priors are sensible and, hence, whether the Bayes factors are meaningful. [...] Personally, I think one of the more effective ways to specify meaningful prior distributions is through prior predictive checking. The authors present posterior predictive checks to assess the models descriptive adequacy, but I think prior predictive checks would be an equally important addition to evaluate the results of the model comparisons.

Reply:

The concern is well-founded. Bayes factors for non-nested models are sensitive to how parameter-space priors translate onto the observable response scale, and identical priors on a and c do not guarantee equivalent implied predictions when the simple and softmax choice rules map those parameters to response probabilities through different mechanisms. We have added prior predictive checks for both choice rules to Tutorial 1.

These checks reveal that the two choice rules do *not* generate equivalent implied predictions under the default priors (see Figure X in the revised manuscript). For the softmax choice rule, the default priors imply a median correct-response proportion of approximately 89% (90% CI: 60–98%) for a representative condition; for the simple choice rule, the same nominal prior specification implies a median of approximately 55% (90% CI: 42–73%). The difference arises because the two choice rules use different link functions — identity for softmax, log for simple — which translate identical-looking parameter-space priors into different response-proportion distributions. This finding validates the reviewer's concern and motivates the practical recommendation we have added to the manuscript: inspecting prior predictive distributions is essential when comparing non-nested models that differ in link structure.

On cell means coding: the implied prior on a condition difference under cell means coding is the convolution of two independent priors on condition-specific activations — sensible in structure, but not directly readable from the prior specification. We have added a footnote in Tutorial 1 > Model Specification explaining this and directing readers to the prior predictive

check as the practical tool for inspecting what the priors imply about expected response proportions in each condition.

Changes:

- Add prior predictive check figure for Tutorial 1 showing implied response proportions under both choice rules
- Add paragraph in Tutorial 1 > Model Evaluation describing the prior predictive procedure and results
- Add footnote in Tutorial 1 > Model Specification explaining cell-means coding and what the default priors imply on the response proportion scale
- Add code chunk showing prior predictive extraction via `posterior_predict()` on prior draws

As an aside, the authors appropriately use bridge sampling to estimate Bayes factors for non-nested model comparisons. Here they report a median estimate across 10 repeated estimation runs, but it would be helpful to also report the variability of the estimates. If I understand correctly the authors use 4000 posterior samples for the estimation, which is generally considered a small number when performing bridge sampling for hierarchical models ("As a rule of thumb, we suggest that testing requires about an order of magnitude more posterior samples than estimation.", 10.18637/jss.v092.i10, p. 12). Few posterior samples may result in highly variable estimates of the marginal likelihoods.

Reply:

The concern is well-grounded. Oberauer, Musfeld, and Aust (2026) demonstrate that bridge sampling estimates for hierarchical GLMs of the complexity used here require approximately 72,000 posterior samples to be reliable — well above the 4,000 samples available from the default 4 chains (× 1,000 post-warmup iterations). We have therefore refit the models used in the choice rule comparison (Tutorial 1) and the custom versus standard model comparison (Tutorial 3) with 20,000 post-warmup iterations per chain (80,000 total). The revised manuscript reports the median Bayes factor and the full min–max range across 10 repetitions for both the default and the long-run sampling regime, illustrating the stability gain directly.

Table ?? summarizes the results for both tutorials. For the choice rule comparison in Tutorial 1, the short-run estimate (median $BF = 4.8 \times 10^7$) and the long-run estimate (median $BF = 6.2 \times 10^7$) agree in direction and order of magnitude. However, the min–max range across 10 repetitions narrows dramatically: from $[3.2 \times 10^7, 7.0 \times 10^7]$ with 4,000 samples to $[5.7 \times 10^7, 6.4 \times 10^7]$ with 80,000 samples. The Tutorial 3 model comparison (custom vs. standard cs) shows a more striking pattern: the short-run median ($BF = 8.1 \times 10^{17}$) is approximately 1,600 times larger than the long-run median ($BF = 5.2 \times 10^{14}$), and the short-run range spans more than three orders of magnitude $[2.1 \times 10^{16}, 7.0 \times 10^{19}]$. With 80,000 samples, the estimate stabilizes to a tight range $[4.7 \times 10^{14}, 5.7 \times 10^{14}]$. In both cases the qualitative conclusion is unchanged — softmax is overwhelmingly favored over simple, and the custom filtering model is overwhelmingly favored over the standard cs model — but the Tutorial 3 results

illustrate why reporting only the short-run estimate would have been misleading.

The hypothesis tests reported throughout the tutorial use a different approach: the Savage-Dickey density ratio via `brms::hypothesis()`, applied to nested comparisons within a single fitted model. Oberauer et al. (2026) show that the Savage-Dickey density ratio scales well across many model parameters and does not require the large sample counts needed for bridge sampling. We have added a brief note distinguishing the two approaches and directing readers to Oberauer et al. (2026) for detailed guidance on when each method is appropriate.

Changes:

- Refit models used for bridge sampling with `iter = 21000` (20,000 post-warmup per chain) in Tutorials 1 and 3
- Report BF as: median *BF* [range: min–max] across 10 repetitions for both sample sizes in both tutorials
- Add explanatory paragraph on bridge sampling sample requirements and the stability diagnostic, citing Oberauer et al. (2026)
- Add brief note distinguishing the Savage-Dickey density ratio (used by `brms::hypothesis()` for nested comparisons) from bridge sampling (used for non-nested model comparisons), citing Oberauer et al. (2026)

In Example 1, the authors present a model comparison between the simple choice rule and the softmax decision rule (e.g., p. 19). They cite recent work that similarly favors softmax over the simple decision rule. As a theoretical justification they cite prior work and discuss that in n -alternative forced choice the softmax rule implies a signal detection theoretic model with Gumble-distributed noise. This implication holds for $n > 2$, whereas for $n = 2$ softmax is equivalent to a signal detection model with logistic distributions (a relatively close approximation to the classic Gaussian distributions). As the package may be applicable to designs with 2-AFC responses, this may be worth briefly mentioning (10.1016/j.jmp.2023.102805). I believe this is relevant also because recent work has found that Gaussian SDT is favored over softmax (Gumble-SDT) in n -alternative forced-choice (10.1016/j.jmp.2023.102805), which I think may also be worth discussing, as it is somewhat at odds with the assertion that "the softmax choice rule is the recommended default based on current evidence" (p. 35). 'bmm' currently only supports simple choice and the softmax decision rule. In contrast to the flexibility in specifying different latent processes, I think it is at least worth mentioning, that specification of the decision rule is less flexible.

Reply:

Both points are well-taken. For 2-AFC tasks, the softmax rule reduces to the logistic function — a close approximation to the Gaussian SDT model with a single discriminability parameter — and we have added a sentence noting this in M3 Framework > Competitive Selection.

On the broader empirical question: Robinson et al. (2023) showed that the softmax scaling parameter varies across the number of alternatives, whereas the signal detection parameter d is stable. This is meaningful evidence against a strong theoretical preference for Gumbel over

Gaussian noise. Meyer-Grant (2025, preprint) has proposed a Gumbel-based extreme-value SDT model for recognition memory with strong empirical support. Taken together, the field has not converged on which competitive selection mechanism is preferable, and we have revised the relevant passage to reflect this ongoing discussion rather than presenting softmax as the recommended default.

Gaussian SDT is not currently implemented in `bmm`, for a practical reason: evaluating the n -AFC Gaussian likelihood requires numerical integration over a multivariate normal distribution. For $n > 2$ there is no closed-form solution, the integration is not natively supported by Stan, and the resulting computational overhead renders hierarchical MCMC estimation impractical. Softmax yields a tractable closed-form likelihood for any number of alternatives. We have revised the text to present softmax as a pragmatic tractable choice rather than a theoretical commitment.

Changes:

- Add note on $n = 2$ softmax equivalence to logistic SDT in M3 Framework > Competitive Selection
- Add sentences acknowledging the ongoing discussion about Gaussian vs. Gumbel SDT, citing Robinson et al. (2023) and Meyer-Grant (2025)
- Add explanation of why Gaussian SDT is not implemented: no closed-form likelihood for $n > 2$; not natively supported by Stan/brms; computationally impractical for hierarchical MCMC
- Soften the "recommended default" phrasing to reflect that the choice of softmax is pragmatic and the theoretical discussion is ongoing

"Researchers can specify M3 of arbitrary complexity for novel task designs, as long as the model remains identified." (p. 10): Is there some guidance on how users can ensure that custom models are identified?

Reply:

We agree. We have added a brief paragraph in Tutorial 3 covering three complementary checks: a conceptual check (each parameter should have a unique empirical signature that no combination of other parameters can reproduce), a practical check (the posterior should be meaningfully narrower than the prior), and a multimodal-posterior check (unimodal marginal posteriors are a necessary but not sufficient condition for identification; multimodal posteriors indicate a problem). These are illustrated for the custom filtering model introduced in Tutorial 3.

Changes:

- Add paragraph in Tutorial 3 > Defining a custom M3 on checking identifiability: (a) conceptual check — each parameter should have a unique empirical signature; (b) practical check — posterior narrower than the prior; (c) multimodal posterior as a diagnostic sign of non-identification

"The model captured the observed response proportions well across all three conditions and five categories." (p. 23): I know this is not the primary point of this example and I suppose a "good" fit as assessed visually from posterior predictions is subjective, but I would say there is noticeable misfit in the "Dist (other)" condition and, to a lesser extent, the "Dist (paired)" condition. As this is addressed by the specification in Tutorial 3, I think it is worth pointing out this misfit here.

Reply:

We agree. The misfit in Dist (other) and Dist (paired) is visible in the figure and was intentionally used to motivate Tutorial 3, but was not flagged explicitly enough. We have added two sentences at the end of Tutorial 2 > Model fitting and evaluation pointing to the residual misfit and forwarding the reader to Tutorial 3.

Changes:

- Add 1–2 sentences in Tutorial 2 > Model fitting and evaluation explicitly noting the residual misfit in the Dist (other) and Dist (paired) categories and forwarding to Tutorial 3

"As in Tutorial 2, r_a and r_c are not identified in the control condition (no distractors present), so we fix them with constant priors on both the fixed effect and the random effect SD" (p. 27): If I understand correctly, in this setup it is no longer arbitrary what values to use for the constant priors, because the resulting coefficient values will be applied across all conditions, is this correct? I assume this is why the prior is set to constant -10 (effectively 0 on the probability scale). If so, I think this should be highlighted here to avoid confusion.

Reply:

The reviewer's reading is correct. The constant value is not arbitrary in the usual sense: it sets the activation baseline from which all condition-specific activations are scaled, so the choice of constant directly shifts all reported condition differences. This means researchers should document the constant they use and note that reported effect sizes are defined relative to it. We have added prose after the t3-priors code chunk making this explicit.

We have also added a note on a related bmm feature not previously described in the tutorial. When a response category has zero response options in a given condition ($n_opts = 0$), bmm automatically suppresses that category by fixing its activation to 100 internally. This handles structurally absent response options and is distinct from the manual constant prior used here, which addresses a parameter that is present in the model but not identified by the available data.

Changes:

- Add prose after the t3-priors code chunk explaining that the constant value is not arbitrary: it sets the baseline for condition contrasts and should be documented
- Add note on bmm's automatic zero- n_opts handling and how it differs from the manual constant prior for non-identification

- Cross-reference briefly in Tutorial 2 where constant (0) is used for f

"As a rule of thumb, if individual-level correlations fall below .7 for a parameter, the design likely needs more trials or a larger sample" (p. 33): What's the justification for this rule of thumb? How did the authors arrive at .7?

Reply:

The $r \geq .7$ threshold is a pragmatic heuristic. It exceeds Cohen's (1988) large-effect convention of $r \geq .50$, providing a margin beyond which individual-difference measurement can be considered adequate for most research purposes. It is not a formally derived criterion, however, and a recovery correlation near .7 warrants caution rather than confidence. We have added a footnote making this explicit and emphasizing that scatter plot inspection is essential alongside any numerical summary.

Changes:

- Add footnote in Tutorial 4 justifying the .7 threshold and noting it is a pragmatic heuristic, not a hard criterion; direct readers to inspect scatter plots

"When parameters are not identified, the posterior will reflect the prior rather than the data, and parameter recovery will fail." (p. 34): Other pathologies, such as label switching (i.e. multimodal distributions), are possible.

Reply:

We agree. Label switching — where parameters serving interchangeable roles are assigned inconsistently across MCMC chains, producing multimodal joint posteriors without necessarily producing unusual marginal posteriors — is a distinct diagnostic concern from non-identification. We have added a brief paragraph in Tutorial 4 describing label switching and directing readers to `mcmc_pairs()` to detect it.

Changes:

- Add brief paragraph in Tutorial 4 > Recovery evaluation on label switching and using `mcmc_pairs()` to detect multimodal joint posteriors

REVIEWER 2

I strongly feel that more words of caution on custom model adjustments and on the need for psychological model validation (no parameter recovery but rather selective influences etc.) are necessary. I worry that the current presentation suggests one can easily assume different sources of activation by adding parameters and different functions but of course the previously suggested functions have been grounded in psychological theory and empirical validation and this would be necessary for any new custom adjustments as well.

Reply:

The concern is well-placed. Parameter recovery establishes that the fitting algorithm can recover known values from data simulated under the model's own assumptions — a computational check, not a psychological one. It says nothing about whether the parameters measure the cognitive processes they are theorized to measure. That form of validation requires theoretical grounding of the activation formulas in cognitive mechanisms, selective influences demonstrations, and convergent validity evidence across tasks and paradigms. The existing M3 formulas have this grounding; a custom specification does not inherit it simply by being implemented in bmm.

We have added a text box at the start of Tutorial 3 making this distinction explicit and outlining the criteria a custom M3 would need to meet to establish psychological validity. The framing is that bmm provides infrastructure for theory development, but the theoretical plausibility and empirical adequacy of any custom specification is the responsibility of the researchers proposing it and the research community evaluating it.

Changes:

- Add text box at start of Tutorial 3 covering: theoretical grounding, identifiability, selective influence, convergent validity, and the role of community evaluation
- Frame bmm explicitly as a workbench for theory development; psychological validity is the responsibility of researchers and the research community, not the software

Relating to validation, the M3 has been developed for working memory and this should be specified throughout. The presentation is currently suggestive that this is a broader memory framework which may also cover episodic memory etc. It has not been developed and validated for such purposes and is not readily applicable to episodic (or semantic, autobiographic etc.) memory tasks and this should be carefully specified throughout.

Reply:

The reviewer is right. M3 has been developed and validated in working memory paradigms, and the tutorial covers working memory applications exclusively. We have revised the relevant passages throughout to reflect this.

We have, however, maintained the claim that the *architecture* of M3 — continuous activation

functions feeding a competitive selection mechanism — is domain-general in a conceptual sense: nothing in the formal structure is specific to working memory. However, applying this architecture in other memory systems would require developing theoretically grounded activation formulas for those domains and establishing that the resulting parameters show the expected selective influences and convergent validity there too. We have revised the Introduction and Summary to make this distinction between architectural generality and current empirical scope explicit throughout.

Changes:

- Revise Introduction paragraph 3 to distinguish the architectural generality of M3 from its current validated scope in working memory
- Add qualifier: extensions to other domains require theoretical formula development and empirical validation in those contexts
- Revise Footnote 1 (MPT comparison) to clarify WM as the current validated scope
- Add sentence to Summary section bounding the tutorial's scope to WM applications

It seems to me that accessibility is the key goal of this tutorial—as the model foundation and formulas have been presented elsewhere and the package development is documented on CRAN. I think this could be improved by avoiding formulas early on as a stand-alone removed from the R package/functions as this might worry away readers not seeing themselves able to implement such formulas. I think it would be best to start with a verbatim description of the model but to leave formulas to the place where it is in parallel explained how they can be accessed and called on by the functions and perhaps some of the formulas are rather for the appendix as they are not newly developed here.

Reply:

We have restructured the M3 Framework section accordingly. In the Activation Formulas subsection, verbal descriptions of each component now precede the formal notation. In the Competitive Selection subsection, the intuitive account and the SDT interpretation are introduced before the equations. The multiplicative activation formula example — which illustrated theoretical flexibility but was not part of the main tutorial workflow — has been moved to a footnote. The goal throughout was to ensure that a reader unfamiliar with formal notation can follow the section without first parsing the equations, and that the equations appear in direct proximity to the R function calls that implement them.

Changes:

- Restructure M3 Framework > Activation Formulas: verbal description first, equation second
- Restructure M3 Framework > Competitive Selection: intuition and SDT interpretation first, equations after
- Move the multiplicative activation formula example to a footnote